NILMBench2026

A Benchmark for Energy Disaggregation

● BuildSys '26 · Banff, Canada

★ Best Paper Candidate

Aayush Kuloor* Anurag Singh* Harsh Dhru* Nipun Batra†

IIT Gandhinagar · * equal contribution † corresponding author

github.com/nilmtk/nilmtk-contrib · sustainability-lab.github.io/nilmbench

What is energy disaggregation?

One aggregate household power signal in → appliance-level estimates out. Fine-grained feedback can cut consumption by up to **15%**.

THE PROBLEM

Lots of models. No reproducible yardstick.

Architectures keep improving — but we can't fairly compare them, and we don't know if they'll **survive deployment**.

What prior benchmarks miss

- **Efficiency ignored** — accuracy reported, but not FLOPs, parameters, or inference time
- **One resolution** — usually 1-min only; ignores utility-scale 15-min
- **Generalization untested** — rarely evaluated across buildings or datasets
- **Fragmented code** — Python 2.7, TensorFlow 1.x; gains conflated with implementation luck

Feature	'14	'19	Ours
Deployability	✗	✗	**uv+Docker**
Models	2	9	**16**
Resolutions	var	1m	**1m & 15m**
Efficiency	✗	✗	✓
Cross-building	✗	✓	✓
Cross-dataset	✗	✗	✓
Stack	Py2.7	TF1	**PyTorch**

NILMBench2026

16	3	2	3	576
models	datasets	resolutions	tasks	configs ×3 runs

- A **reproducible, deployment-aware** benchmark across accuracy, efficiency & generalization
- We **modernize NILMTK**: every legacy model re-implemented in **PyTorch** under one API
- One-command reproducibility with `uv` + **Docker**, and **+5** modern architectures

Reproduce every result in 3 commands

```
# install the modernized stack — 16 models, one PyTorch API
uv pip install git+https://github.com/nilmtk/nilmtk-contrib.git
# …or a pinned, GPU-ready container
docker run --gpus all ghcr.io/enfuego27826/nilmtk-contrib:latest bash
```

```
from nilmtk.api import API
from nilmtk_contrib.disaggregate import NILMFormer, Seq2Point, TCN

experiment['methods'] = {'NILMFormer': NILMFormer({'n_epochs': 50}), ...}
results = API(experiment)     # trains, tests & scores every model
```

Adding a model = subclass `Disaggregator`. Adding a metric = one function. Same harness for everyone.

Datasets — two countries, two grids

Dataset	Country	Buildings	Duration	Appliances
REDD	USA (110 V)	6	3–19 days	10–20
UK-DALE	UK (230 V)	5	655 days	5–54
REFIT	UK (230 V)	20	2 years	9–21

Six appliances spanning NILM difficulty: `Fridge` `Microwave` `Kettle` `Washing Machine` `Dish Washer` `Television` · excluded: single-building (AMPds, iAWE...) & pay-walled (PecanStreet)

16 architectures, 4 families

RECURRENT & HYBRID

RNN · WindowGRU · **ConvLSTM** · RNN-Attn · RNN-Attn-Cl

TRANSFORMER-BASED

BERT · **Reformer** · **NILMFormer**

FULLY CONVOLUTIONAL

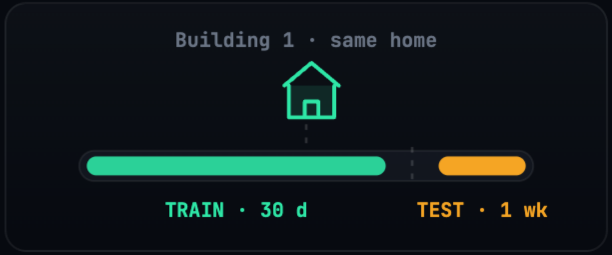
Seq2Point · Seq2Seq · **TCN** · ResNet · ResNet-Cl

SPECIALIZED NILM

DAE · **MSDC**

green = the **5 architectures we add** in NILMBench2026 (TCN, ConvLSTM, MSDC, Reformer, NILMFormer)

Three tasks — increasing realism

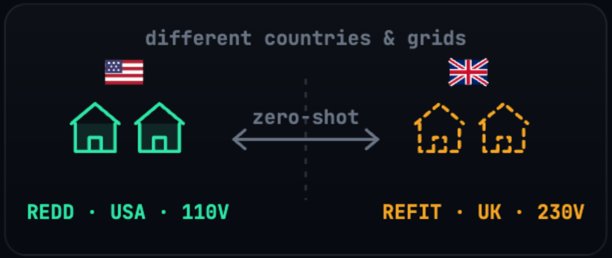


T1 · Intra-Building

Same home, train past → test held-out week. Best case.

T2 · Cross-Building

Unseen home, **same country**. The realistic test.



T3 · Cross-Dataset

Zero-shot, **different country & grid**. Hardest.

FINDING 1

No single model wins.

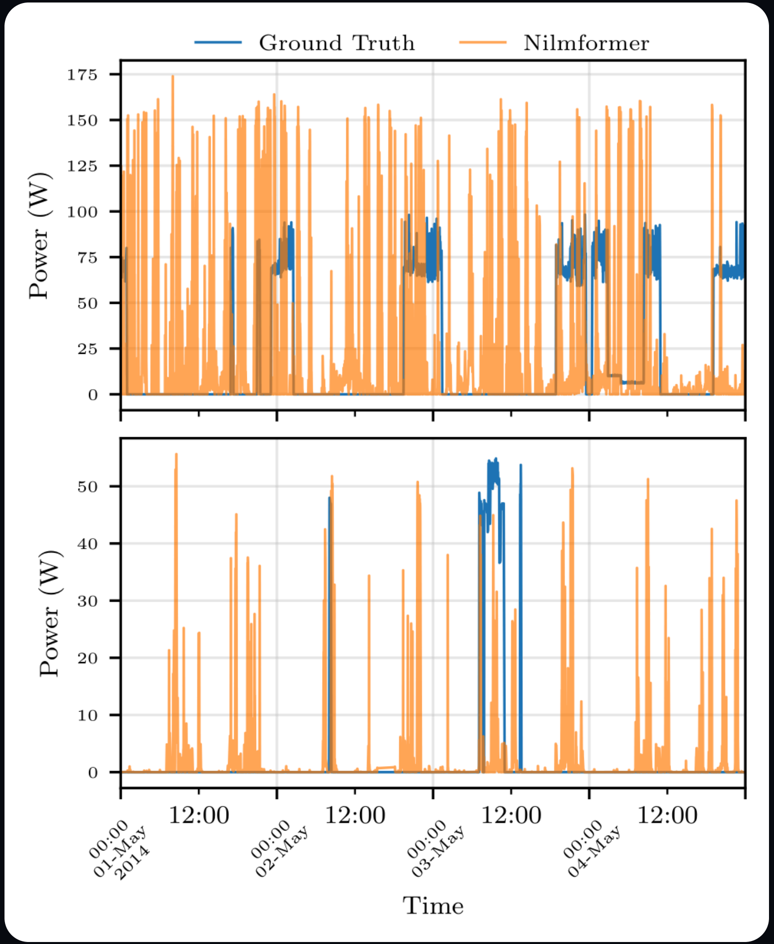
Performance is highly **context-dependent** — the best architecture changes with appliance type and time resolution.

- **1-min:** deep CNNs (Seq2Point, TCN) capture sharp activation transients
- **15-min:** averaging erases detail — NILMFormer's attention & exogenous priors win
- CNNs excel at sparse bursts; Transformers handle complex multi-state loads

Finding 2 — generalization is the wall

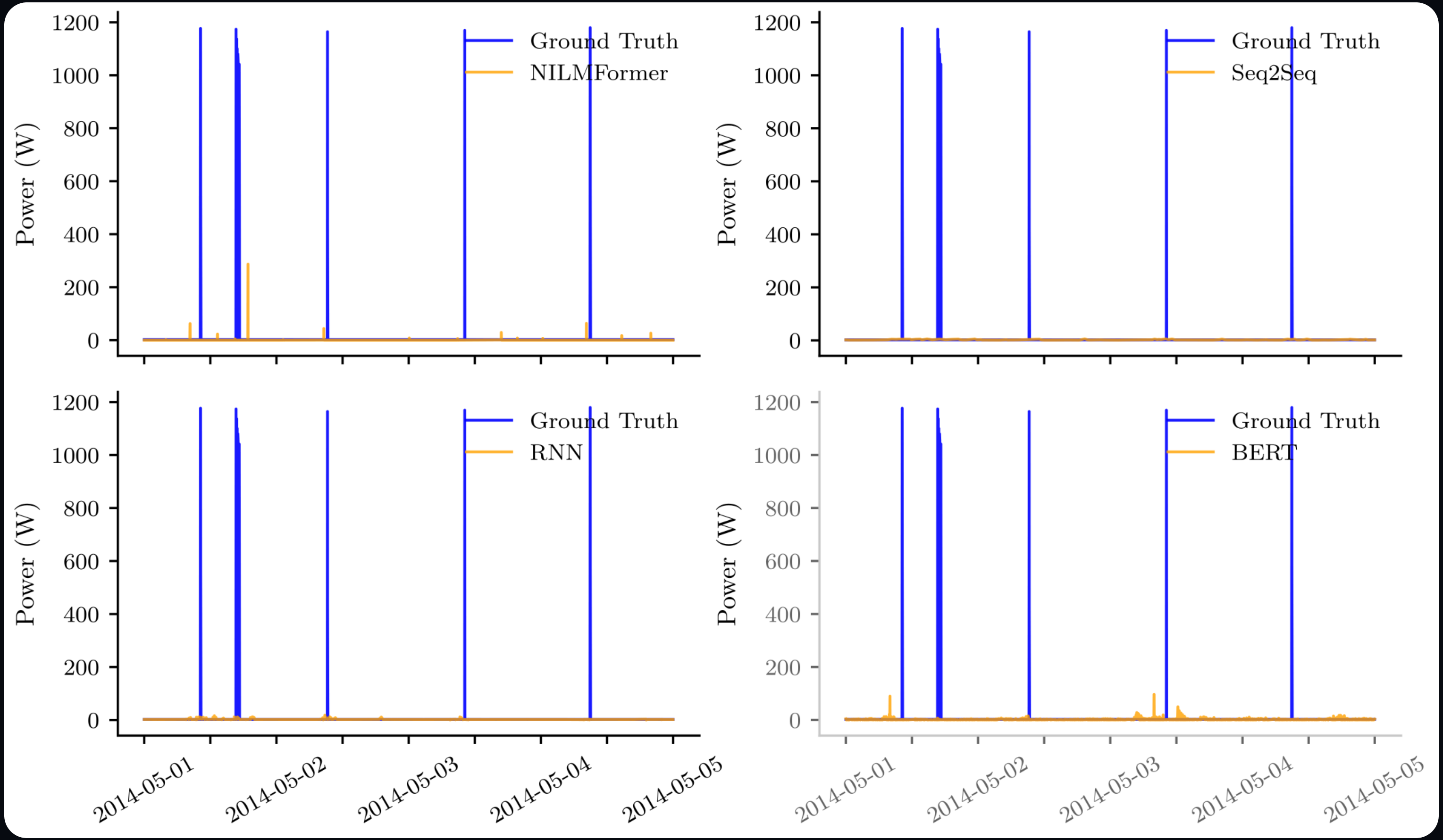
- Accuracy **collapses** from T1 → T2 → T3, symmetrically in both transfer directions
- Models memorize **one device's electrical fingerprint** — not the abstract appliance
- Right: NILMFormer on a TV — tracks the trained set (bottom), **fails** on an unseen TV (top)

> The drop from same-building to unseen-building is the **core barrier to real-world NILM**.



Finding 3 — MAE is misleading

On sparse loads, a model scores a low MAE by always predicting **"off"** — while missing **every** activation.



REFIT microwave (cross-building): four models miss every 1200 W spike, yet look "accurate" by MAE. **Event metrics (F1) are essential.**

Finding 4 — efficiency ≠ accuracy

The accuracy–compute trade-off is **non-monotonic**. Architectural inductive bias beats raw compute.

- **TCN** — just **69K params** — rivals the heavyweight NILMFormer on cross-dataset tasks
- Smallest models consistently **under**-perform; biggest aren't the best
- Real deployment likely needs an **ensemble** of specialists

Model	MAE↓	GFLOPs	Params
NILMFormer	**15.4**	135	383K
Seq2Point	18.4	72	3.6M
TCN	20.8	**20**	**69K**
BERT	23.7	**5.4**	803K

T2 cross-building (UK-DALE), mean MAE.

Four takeaways

🏆 No single model wins

Best architecture depends on the appliance signature — CNNs for bursts, Transformers for multi-state.

🧱 Generalization is the hurdle

Models fail on unseen buildings & datasets — the barrier to adoption.

🎭 MAE is misleading

Predicting “off” hides missed events. Report F1 for sparse loads.

⚖️ Efficiency ≠ accuracy

Non-monotonic: a 69K-param TCN rivals heavyweights.

A LIVING BENCHMARK

Built to become NILM's ImageNet.

ImageNet moved vision. GLUE moved language. NILM has never had a shared, sealed, generalization-first leaderboard — until now.

- **Add a model** in one class · **add a metric** in one function · same frozen harness
- Submit to a public, OOD-first **leaderboard** — currently led by NILMFormer (T2 MAE 15.4)
- Toward an annual **NILM Challenge** in the spirit of the KDD Cup

Conclusion

NILMBench2026 evaluates **16 models × 3 datasets × 2 resolutions** on accuracy, efficiency & generalization — and ships the reproducible platform to keep doing so.

- **Generalization is the prerequisite** for deploying NILM at scale — not marginal accuracy
- No universal model; MAE misleads on sparse loads; efficiency is decoupled from accuracy
- **Next:** domain adaptation · self-supervised pre-training · adaptive denormalization · community leaderboards

Thank you

Generalization is the wall. Let's tear it down — together.

Aayush Kuloor · Anurag Singh · Harsh Dhru · Nipun Batra

nipun.batra@iitgn.ac.in · IIT Gandhinagar

📄 `sustainability-lab.github.io/nilmbench`

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